

Milestone 01 — Download and Prepare Covariates

Structured Acquisition and QA of Inputs for Regional Skew Estimation

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July 22, 2025

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1 Overview of v1.4

This document outlines the setup, documentation, and reproducibility scaffolding established for **Milestone v1.4**, focused on acquiring and validating spatial covariates for regional skew modeling.

This milestone builds on `v0.3-structure-refactor`.

1.0.1 Summary

Initiate acquisition, validation, and preparation of climate, terrain, and location-based covariates for use in regional skew estimation models.

1.1 Goals

Update the covariate source inventory:

- Write or refactor download scripts
- Document file sources and formats
covariate_source_inventory.md
- Prepare spatial data
spatial_data_preparation_checklist.md
- Acquire and clean metadata
metadata
- Apply version control and tagging

2 Project Structure

```
FFA_regional-skew/
.gitignore           # Prevents sensitive/local files from being pushed
arctgis_project/     # ArcGIS Pro .aprx project and supporting layers
CHANGELOG.Rmd        # Human-readable changelog (semantic versioning)
data/                # Raw, processed, meta, intermediate, and QA data
docs/                # Documentation, workflow notes, metadata, checklists
FFA_regional-skew.Rproj # RStudio project file (keep in root)
notebooks/           # Ad hoc .Rmd/.qmd experiments (empty or minimal)
notes/               # Team notes, meeting logs, brainstorm (convert to .md/.qmd)
R/                   # Analysis scripts (milestone-organized)
  01_download/       # Download raw data
    _latex_preamble.tex # LaTeX preamble for PDF reports
    01a_download_ecoregions.R # Download U.S. ecoregions
    01b_download_USGS_gage_data.R # Download raw peak flow site data
    01c_download_usgs_gage_metadata.R # Download USGS site metadata
    01d_download_usgs_peakflow_data.R # Download peak flow data
    01e_filter_usgs_peakflow_data.R # Filter peak flow data
    01f_download_nhdplus_v21_flowlines.R
    01g_download_nhdplus_hr_catchments.R
    01h_download_nhdplus_hr_flowlines.R
    01j_download_koppen_geiger_climate.R
    01k_download_plant_hardiness_zone_map.R
    01l_download_prism_climate.R
    01m_download_nlcd_2016.R
    01n_download_ned.R
    01o_download_modis_2016.R
    01p_download_statsgo2.R
  02_clean_validate/ # QA scripts and validation checks
  03_covariates/     # Covariate extraction: climate, terrain, land cover
  04_modeling/       # Statistical modeling (GAMs, Elastic Net, etc.)
  06_eval/           # Model diagnostics, residuals, validation
  utils/             # Reusable functions (organized by domain)
    metadata/
    spatial/
    qaqc/
```

```

    paths/
    plotting/
README.md          # GitHub-readable overview and navigation aid
README.Rmd         # Editable RMarkdown version with richer formatting
reports/          # Knitted .Rmd/.qmd milestone reports
                  # Next: consider `reports/final/` and `reports/draft/`
results/          # Manuscript-ready figures, models, and outputs
  maps/
  tables/
  models/
  posterdown/     # Poster files and assets
  slides/         # Slide decks and presentation visuals
sandbox/          # Staging area for review

```

3 v1.4 – Download and Prepare Covariates

4 v1.4 Tasklist

Step	Task	Status
1.4.1	Refine the Covariate Inventory	[X]
1.4.2	Update folder structure	[X]
1.4.3	Create / update download scripts for vector data	[X]
1.4.4	Create downloads script for raster covariates	[X]
1.4.5	QAQC for downloads	[X]
1.4.6	Knit milestone and data dictionary .Rmd files to PDF	[X]
1.4.7	Document changes in 01_download README.Rmd	[X]
1.4.8	Commit and tag v1.4-download-scripts	[X]

5 Changelog Format: Per-Step Narrative

5.0.1 Step 1.4.1 — Refine the Covariate Inventory

Actions:

- Finalized covariate list by domain (climate, terrain, land cover)
- Documented filenames, source URLs, formats, versioning, and priority
- Created reusable QA checklists and source tracking templates

Reason (Before):

The project lacked a centralized, versioned inventory of covariates. Metadata was fragmented across exploratory scripts with no QA framework.

Result (After):

- Created docs/covariate_source_inventory.md
- Created docs/spatial_data_preparation_checklist.md for 12-step QA
- Updated data/meta/covariates_metadata_schema.csv
- Standardized metadata for reproducibility and audit tracking

5.0.2 Step 1.4.2 — Update Folder Structure for utils/ and data/

Actions:

- Backed up project before restructuring
- Created new subfolders under data/: raw/, processed/, meta/, log/, intermediate/
- Created new utility folders in R/utils/ for:
 - metadata/
 - spatial/
 - qaqc/
 - paths/
 - plotting/

Reason (Before):

Functions were embedded in monolithic scripts or scattered, limiting reuse and debugging.

Result (After):

Utility functions are modular, testable, and purpose-scoped. Folder structure supports test-driven development and script clarity.

5.0.3 Step 1.4.3 — Create / Update Download Scripts for Vector Data

Actions:

- Updated ecoregions, USGS gage, NHDPlus, and STATSGO2 download scripts
- Harmonized all vector outputs to EPSG:5070
- STATSGO2 spatial queries now chunked and geometry-validated

Reason (Before):

Inconsistent file structures, missing scripts, and fragile STATSGO2 queries.

Result (After):

All vector datasets reproducibly downloaded, spatially filtered to Great Plains AOI, and exported to data/raw/ and data/processed/. STATSGO2 joins are verified by mukey.

5.0.4 Step 1.4.4 — Create Download Scripts for Raster Covariates

Actions:

- Created modular download and processing scripts for:
 - Köppen-Geiger
 - USDA Plant Hardiness Zones
 - PRISM climate normals (via {prism})
 - NLCD 2016
 - NED elevation and slope (via {terra})
 - MODIS NDVI 2016 (via {MODISTools})
- Standardized reprojection (EPSG:5070), clipping, and export

Reason (Before):

Raster layers were manually downloaded, inconsistently named, or lacked projections.

Result (After):

All rasters downloaded, cleaned, and saved to `data/processed/` with consistent filenames, formats, and CRS. Metadata written to `data/meta/`.

5.0.5 Step 1.4.5 — QAQC for Downloads

Actions:

- Validated CRS, resolution, and extent across all vector and raster layers
- Created summary metadata export including layer name, type, CRS, resolution, extent, and notes
- Tagged .Rmd files with `{r name, eval=}` chunk headers for reproducibility
- Documented internal workflow in:
 - `git_changelog_workflow_reference.Rmd`
 - `spatial_data_preparation_checklist.Rmd`

Reason (Before):

Unvalidated spatial layers posed a risk of alignment errors. Documentation was scattered or incomplete.

Result (After):

All layers validated for modeling compatibility. Scripts are now reproducible, documented, and version-controlled. Project directories are aligned with workflow logic and audit-ready.

5.0.6 Steps 1.4.6 to 1.4.8

Actions

This update focused on documenting and standardizing the spatial data download scripts housed in `R/01_download/`. A new `README.Rmd` was created to summarize the purpose, inputs, outputs, and dependencies of each script. Workflow summaries and milestone cross-references were added to all major download scripts to improve clarity and reproducibility.

Reason (Before):

Script headers were standardized to include consistent metadata and dependency blocks. Export paths and file-saving logic were refined for transparency, and minor cleanup was applied to ensure consistency across the download workflow.

Result (After):

These updates improve maintainability and prepare the scripts for downstream integration in covariate processing and modeling steps.